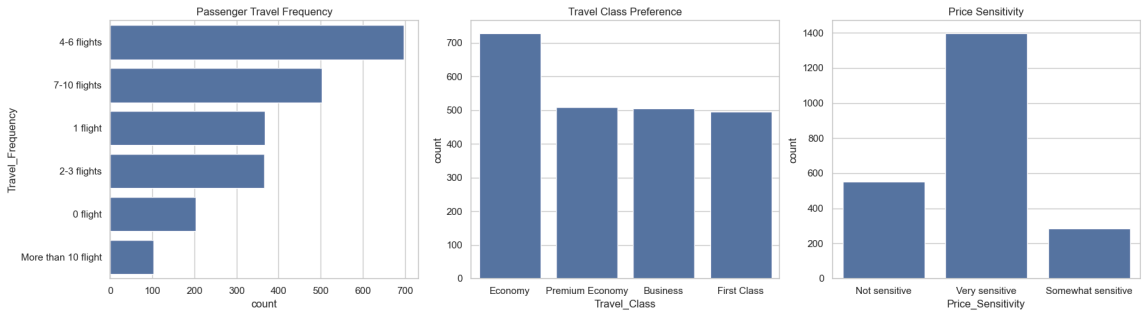
Airline Data Analysis Report

Comprehensive analysis of passenger demographics, loyalty behavior, and customer sentiment.

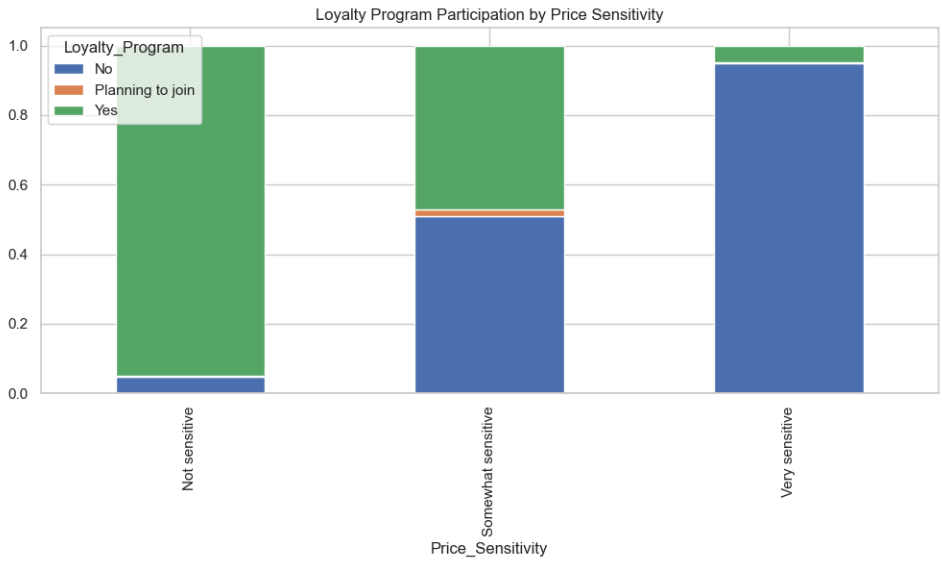
1. Passenger Demographics

Observation: Most travelers fly 4-6 times/year. Economy is dominant, but Premium classes show significant volume.



2. The Loyalty-Price Disconnect

CRITICAL INSIGHT: Loyalty is almost exclusive to price-insensitive users. The program fails to capture the price-sensitive mass market.

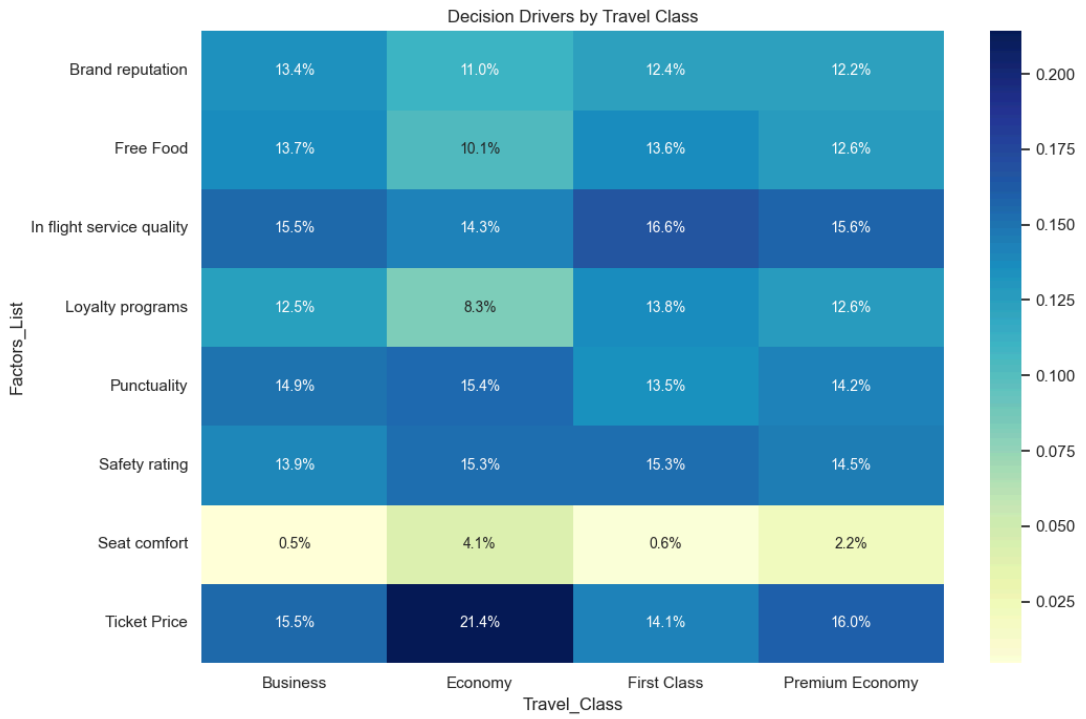


3. Decision Drivers by Class

We analyzed what influences a passenger's decision based on their cabin class. The Heatmap below shows a clear split:

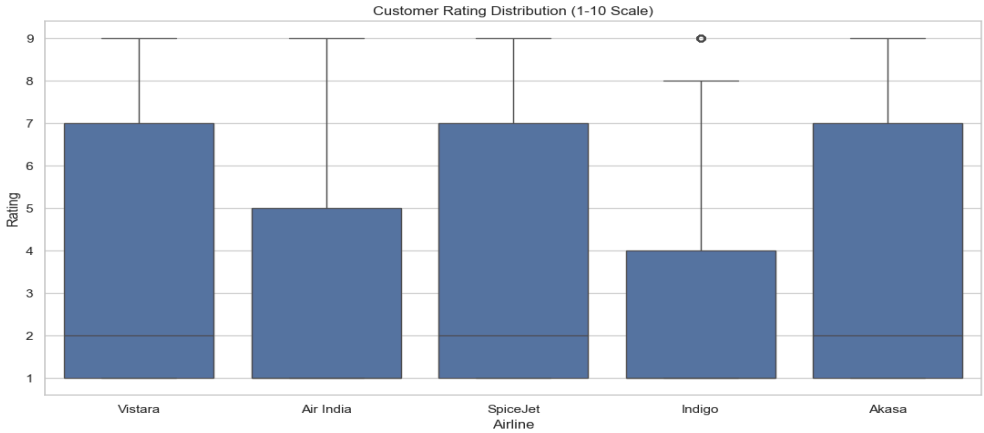
- Economy Flyers are driven by Ticket Price.

- First/Business Flyers are driven by Brand Reputation and Punctuality.



4. Customer Rating Consistency

The boxplot below shows the spread of ratings (1-10). Larger boxes indicate inconsistent service quality.

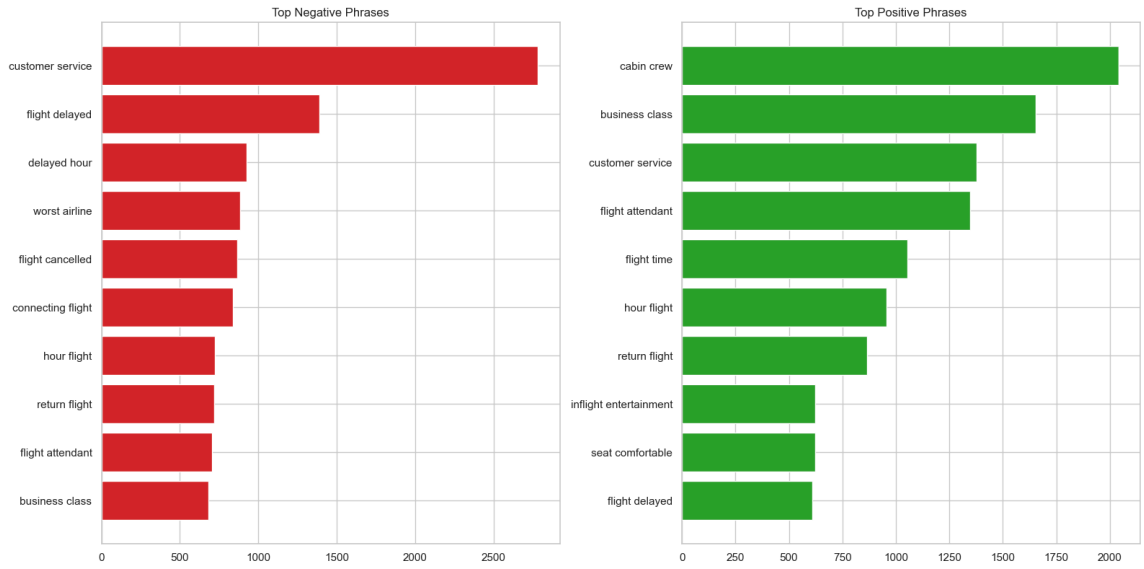


5. Top Customer Phrases

Frequency analysis of 2-word phrases in reviews:

- Negative: dominated by 'Customer Service' and 'Connecting Flight'.

- Positive: dominated by 'Good Food' and 'Cabin Crew'.



6. Strategic Recommendations

1. SEGMENTED MARKETING: Market 'Low Fares' to Economy and 'Punctuality' to Business.

2. REWARD OVERHAUL: Create a 'Lite' tier for leisure travelers who want cashback, not lounge access.

3. SERVICE RECOVERY: Focus training on 'Customer Service' interactions, as this is the primary driver of negative reviews.